



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Artificial Intelligence and Data Science Department

Subject Name	Artificial Intelligence	A.Y	2026-27
Subject Code	303105307	Semester	5 th
Chapter-1			
Sr No	Question	COs	B.T
1	What is AI and major areas of AI?		
2	What are some challenges faced by AI systems?		
3	What are some key characteristics of well-defined problems for AI.		
4	Give two examples of AI applications in e-commerce.		
5	What are the properties of an internal knowledge representation?		
6	What are some challenges in designing search algorithms for AI problems?		
7	How can AI problems be viewed as state space search problems?		
8	What is the purpose of knowledge representation in AI?		
9	What are the main components of an intelligent agent architecture?		
10	What is an intelligent agent in AI?		
11	Discuss the key characteristics that differentiate intelligent behavior in machines from simple automation.		
12	Explain three commonly used AI techniques.		
13	Briefly outline the history of Artificial Intelligence research, highlighting some key milestones.		
14	Describe three major challenges faced by current AI systems.		
15	Explain three key characteristics that make a problem well-suited for solving with AI techniques.		
16	Differentiate between two types of intelligent agents based on their decision-making capabilities.		
17	Discuss how each component contributes to the agent's ability to interact with its environment.		
18	Describe two common techniques for representing knowledge within a machine learning system.		
19	Discuss two potential future directions for research in Artificial Intelligence.		
20	Describe two main challenges faced when designing search algorithms for AI problems.		